

Physics Midterm 2

Unit 4

- 1) B: 45 rpm, 4.7 radians sec^{-1}
- 2) D: 2.2 ms^{-2}
- 3) D: $w \rightarrow \text{low}$
- 4) B: 0.53 N
- 5) C: 42 ms^{-1}

Unit 5

- 1) D: 200 N m, 37 degrees

2) a. $w = F \cdot d$

$$w = 1000 \text{ N} \cdot 0.1 \text{ m}$$

$$w = 100 \text{ J}$$

b. $\Delta K = 100 \text{ J}$

c. $\Delta K = \frac{1}{2} m v^2 \rightarrow v = \sqrt{\frac{200 \text{ J}}{120}}$

$$v = 36.5 \text{ m/s}$$

d. $R = \frac{v^2}{g} \sin^2(20)$

$$= \frac{(36.5)^2}{9.81}$$

3) a. $F_k = \mu_k \cdot N$

$$N = 40 \times 9.81 = 392.4 \text{ N}$$

$$w = F_k \cdot d$$

$$F_k = .05 \times 392.4 =$$

$$w = 19.62 \times 40$$

$$= 19.62 \text{ N}$$

$$w = 784.8 \text{ J}$$

$$R = 135.9 \text{ m}$$

b. $p = \frac{w}{t} = \frac{784.8}{30 \text{ s}}$

$$p = 26.16 \text{ J/s}$$

4) D: \$2.33

Unit 6

1) C: 0 m/s

$$2) v_f = \frac{p_i}{m_1 + m_2} = \frac{0}{m_1 + m_2} = 0 \quad \boxed{v_f = 0 \text{ m/s}}$$

$$3) a. \text{impulse} = F \cdot \Delta t = 4000 \times 0.20$$

$$\boxed{J = 800 \text{ N}\cdot\text{s}}$$

$$b. 1800 = 200 \cdot (v_f - 2.80)$$

$$800 = 200 v_f - 560$$

$$\frac{200 v_f}{200} = \frac{1360}{200}$$

4) C: Elastic

$$\boxed{v_f = 6.80 \text{ m/s}}$$

$$5) a. (30,000)(0.85) + (110,000)(0) = (30,000 + 110,000) v_f$$

$$v_f = \frac{25,500}{140,000} = 0.182 \text{ m/s} \quad \boxed{v_f = 0.182 \text{ m/s}}$$

$$b. KE = \frac{1}{2} m v^2$$

$$= \frac{1}{2} (30000) (0.850)^2$$

$$KE = 10837.50 \text{ J (start)}$$

$$KE = \frac{1}{2} (140000) (0.182)^2$$

$$KE = 2318.68 \text{ J (end)}$$

$$\text{start } 10837.50 \text{ J}$$

$$\text{end } - 2318.68 \text{ J}$$

$$\boxed{8518.82 \text{ J lost}}$$

Unit 7

1) A: Remains motionless

$$2) w = m \cdot g = 20 \times 9.81 = 196.2 \text{ N} = F_1 + F_2$$

$$F_1 = \frac{98.1}{2}$$

$$\boxed{\vec{F}_1 = 49.05 \text{ N}}$$

$$196.20$$

$$- 49.05$$

$$\boxed{\vec{F}_2 = 147.15 \text{ N}}$$

Unit 8

$$\begin{aligned} 1) a. \tau &= \frac{1}{2}MR^2 \\ &= \frac{1}{2}(1)(0.01) \\ \tau &= 0.0032 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

$$\begin{aligned} b. \tau &= I \cdot \alpha \\ &= 0.0032 \cdot 2 \\ \tau &= 0.0064 \text{ N} \cdot \text{m} \end{aligned}$$

$$2) \vec{a} = -0.20 \quad v_x = 0.20 \quad F_x = 12 \quad F_y = 0$$

$$\begin{aligned} \vec{\tau} &= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}((-0.20)(0) - (0.20)(12)) \\ \vec{\tau} &= -2.4 \hat{k} \text{ N} \cdot \text{m} \end{aligned}$$

$$\begin{aligned} b. \tau_z &= r_x F'_y - r_y F'_x \\ 2.4 &= (-0.20)(F'_y) - (0.20)(F'_x) \end{aligned}$$

$$\begin{aligned} 2.4 &= (-0.20)(F'_y) \\ F'_y &= -\frac{2.4}{-0.20} = 12 \text{ N} \end{aligned}$$

$$\vec{F}_{\text{new}} = 0 \hat{i} + 12 \hat{j} \text{ N} = 12 \hat{j} \text{ N}$$